Supplementary Material to the paper entitled: "The Overlooked Metric: Best Case Performance of Efficient Non-dominated Sort"

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1 Best Case Analysis of ENS-SS

The number of dominance comparisons in case 1 is calculated using Eq. (1).

$$
\begin{aligned}
\mathtt{dC}_{\mathtt{SS}-1} &= \sum_{k=1}^{K}\left[\frac{1}{2}n_k^2 + (k-1)n_k\right] - \frac{1}{2}N \\
&= \sum_{k=1}^{X}\left[\frac{1}{2}\left\{X-(k-1)\right\}^2 + (k-1)\left\{X-(k-1)\right\}\right] - \frac{1}{2}N \\
&= \frac{1}{2}\sum_{k=1}^{X}\left\{X-(k-1)\right\}^2 + \sum_{k=1}^{X}(k-1)\left\{X-(k-1)\right\} - \frac{1}{2}N \\
&= \frac{1}{2}\sum_{k=1}^{X}\left[X^2 + (k-1)^2 - 2X(k-1)\right] + \sum_{k=1}^{X}X(k-1) - \sum_{k=1}^{X}(k-1)^2 - \frac{1}{2}N \\
&= \frac{1}{2}\sum_{k=1}^{X}X^2 + \frac{1}{2}\sum_{k=1}^{X}(k-1)^2 - \sum_{k=1}^{X}X(k-1) + \sum_{k=1}^{X}X(k-1) - \sum_{k=1}^{X}(k-1)^2 - \frac{1}{2}N \\
&= \frac{1}{2}X^3 - \frac{1}{2}\sum_{k=1}^{X}(k-1)^2 - \frac{1}{2}N \\
&= \frac{1}{2}X^3 - \frac{1}{12}X(X-1)(2X-1) - \frac{1}{2}N \\
&= \frac{1}{2}X^3 - \frac{1}{12}X\left(2X^2-3X+1\right) - \frac{1}{2}N \\
&= \frac{1}{2}X^3 - \frac{1}{6}X^3 + \frac{1}{4}X^2 - \frac{1}{12}X - \frac{1}{2}N \\
&= \frac{1}{3}X^3 + \frac{1}{4}X^2 - \frac{1}{12}X - \frac{1}{2}N
\end{aligned} \tag{1}
$$

The number of dominance comparisons in case 3 is calculated using Eq. (2).

$$\begin{aligned}
\mathtt{dC}_{\mathtt{SS}-3} &= \sum_{k=1}^{K}\left[\frac{1}{2}n_k^2 + (k-1)n_k\right] - \frac{1}{2}N \\
&= \sum_{k=1}^{X-\delta}\left[\frac{1}{2}(X+z-(k-1))^2 + (k-1)(X+z-(k-1))\right] - \frac{1}{2}N \\
&= \underbrace{\frac{1}{2}\sum_{k=1}^{X-\delta}(X+z+1-k)^2}_{A_{\mathtt{SS-3}}} + \underbrace{\sum_{k=1}^{X-\delta}(k-1)(X+z+1-k)}_{B_{\mathtt{SS-3}}} - \frac{1}{2}N
\end{aligned} \tag{2}$$

Now we first compute $A_{\mathtt{SS-3}}$ and $B_{\mathtt{SS-3}}$.

$$\begin{aligned}
A_{\mathtt{SS-3}} &= \frac{1}{2}\sum_{k=1}^{X-\delta}(X+z+1-k)^2 \\
&= \frac{1}{2}\sum_{k=1}^{X-\delta}\left[(X+z+1)^2 + k^2 - 2k(X+z+1)\right] \\
&= \frac{1}{2}(X+z+1)^2\sum_{k=1}^{X-\delta}1 + \frac{1}{2}\sum_{k=1}^{X-\delta}k^2 - (X+z+1)\sum_{k=1}^{X-\delta}k \\
&= \frac{1}{2}(X+z+1)^2(X-\delta) + \frac{1}{12}(X-\delta)(X-\delta+1)(2X-2\delta+1) - \frac{1}{2}(X+z+1)(X-\delta)(X-\delta+1) \\
&= \frac{1}{2}(X+z+1)^2(X-\delta) + \frac{1}{12}(X-\delta)(X-\delta+1)\left[2X-2\delta+1-6X-6z-6\right] \\
&= \frac{1}{2}(X+z+1)^2(X-\delta) + \frac{1}{12}(X-\delta)(X-\delta+1)(-4X-2\delta-6z-5) \\
&= \frac{1}{2}(X+z+1)^2(X-\delta) - \frac{1}{12}(X-\delta)(X-\delta+1)(4X+2\delta+6z+5)
\end{aligned} \tag{3}$$

$$\begin{aligned}
B_{\mathtt{SS-3}} &= \sum_{k=1}^{X-\delta}(k-1)(X+z+1-k) \\
&= \sum_{k=1}^{X-\delta}\left[(X+z+1)k - (X+z+1) - k^2 + k\right] \\
&= (X+z+2)\sum_{k=1}^{X-\delta}k - (X+z+1)\sum_{k=1}^{X-\delta}1 - \sum_{k=1}^{X-\delta}k^2 \\
&= \frac{1}{2}(X+z+2)(X-\delta)(X-\delta+1) - (X+z+1)(X-\delta) - \frac{1}{6}(X-\delta)(X-\delta+1)(2X-2\delta+1) \\
&= \frac{1}{6}(X-\delta)(X-\delta+1)\left[3X+3z+6-2X+2\delta-1\right] - (X+z+1)(X-\delta) \\
&= \frac{1}{6}(X-\delta)(X-\delta+1)(X+2\delta+3z+5) - (X+z+1)(X-\delta)
\end{aligned} \tag{4}$$

Now put the values of $A_{\mathtt{SS-3}}$ and $B_{\mathtt{SS-3}}$ in Eq. (2) to obtain the number of dominance comparisons in case 3 using Eq. (5).

$$\begin{aligned}
\mathtt{dC}_{\mathtt{SS}-3} &= A_{\mathtt{SS-3}} + B_{\mathtt{SS-3}} - \frac{1}{2}N \\
&= \frac{1}{2}(X+z+1)^2(X-\delta) - \frac{1}{12}(X-\delta)(X-\delta+1)(4X+2\delta+6z+5) +
\end{aligned}$$

$$
\begin{aligned}
&\frac{1}{6}(X-\delta)(X-\delta+1)(X+2\delta+3z+5)-(X+z+1)(X-\delta)-\frac{1}{2}N\\
&=\frac{1}{2}(X+z+1)^2(X-\delta)-(X+z+1)(X-\delta)-\frac{1}{12}(X-\delta)(X-\delta+1)(4X+2\delta+6z+5)+\\
&\frac{1}{6}(X-\delta)(X-\delta+1)(X+2\delta+3z+5)-\frac{1}{2}N\\
&=\frac{1}{2}(X+z+1)(X-\delta)\left[X+z+1-2\right]-\frac{1}{12}(X-\delta)(X-\delta+1)\left[4X+2\delta+6z+5-2X-4\delta-6z-10\right]-\frac{1}{2}N\\
&=\frac{1}{2}(X+z+1)(X-\delta)(X+z-1)-\frac{1}{12}(X-\delta)(X-\delta+1)(2X-2\delta-5)-\frac{1}{2}N\\
&=\frac{1}{2}\left[(X+z)^2-1\right](X-\delta)-\frac{1}{12}\left[(X-\delta)^2+(X-\delta)\right](2X-2\delta-5)-\frac{1}{2}N\\
&=\frac{1}{2}\left[X^2+2Xz+(z^2-1)\right](X-\delta)-\frac{1}{12}\left[X^2-(2\delta-1)X+(\delta^2-\delta)\right][2X-(2\delta+5)]-\frac{1}{2}N\\
&=\frac{1}{2}\left[X^3+(2z-\delta)X^2+(z^2-2z\delta-1)X-\delta(z^2-1)\right]-\frac{1}{12}\left[2X^3-(2\delta+5)X^2-2(2\delta-1)X^2+\right.\\
&\left.(2\delta-1)(2\delta+5)X+2(\delta^2-\delta)X-(\delta^2-\delta)(2\delta+5)\right]-\frac{1}{2}N\\
&=\frac{1}{12}\left[6X^3+6(2z-\delta)X^2+6(z^2-2z\delta-1)X-6\delta(z^2-1)\right]-\frac{1}{12}\left[2X^3-(2\delta+5)X^2-2(2\delta-1)X^2+\right.\\
&\left.(2\delta-1)(2\delta+5)X+2(\delta^2-\delta)X-(\delta^2-\delta)(2\delta+5)\right]-\frac{1}{2}N\\
&=\frac{1}{12}(6X^3-2X^3)+\frac{1}{12}\left[12z-6\delta+2\delta+5+4\delta-2\right]X^2+\\
&\frac{1}{12}\left[(6z^2-12z\delta-6)-(2\delta-1)(2\delta+5)-(2\delta^2-2\delta)\right]X+\frac{1}{12}\left[-6\delta(z^2-1)+(\delta^2-\delta)(2\delta+5)\right]-\frac{1}{2}N\\
&=\frac{1}{12}4X^3+\frac{1}{12}(12z+3)X^2+\frac{1}{12}\left[6z^2-12z\delta-6-4\delta^2-8\delta+5-2\delta^2+2\delta\right]X+\\
&\frac{1}{12}\left[(\delta^2-\delta)(2\delta+5)-6\delta(z^2-1)\right]-\frac{1}{2}N\\
&=\frac{1}{12}4X^3+\frac{1}{12}(12z+3)X^2+\frac{1}{12}\left[6z^2-12z\delta-6\delta^2-6\delta-1\right]X+\\
&\frac{1}{12}\left[(\delta^2-\delta)(2\delta+5)-6\delta(z^2-1)\right]-\frac{1}{2}N\\
&=\frac{1}{3}X^3+\frac{1}{4}(4z+1)X^2-\frac{1}{12}\left[6\delta^2+(12z+6)\delta-(6z^2-1)\right]X+\frac{1}{12}\left[(\delta^2-\delta)(2\delta+5)-6\delta(z^2-1)\right]-\frac{1}{2}N\\
&=\frac{1}{3}X^3+\frac{1}{4}(4z+1)X^2-\frac{1}{12}\left[6\delta^2+6(2z+1)\delta-(6z^2-1)\right]X+\frac{1}{12}\left[(\delta^2-\delta)(2\delta+5)-6\delta(z^2-1)\right]-\frac{1}{2}N\\
&=\frac{1}{3}X^3+\frac{1}{4}(4z+1)X^2-\frac{1}{12}\left[6\delta^2+6(2zX-\delta^2)-(6z^2-1)\right]X+\\
&\frac{1}{12}\left[(\delta^2-\delta)(2\delta+5)-6\delta(z^2-1)\right]-\frac{1}{2}N\qquad\text{Replace the value of } (2z+1)\delta \text{ from Eq. (19) in the paper}\\
&=\frac{1}{3}X^3+\frac{1}{4}(4z+1)X^2-\frac{1}{12}\left[12zX-(6z^2-1)\right]X+\frac{1}{12}\left[(\delta^2-\delta)(2\delta+5)-6\delta(z^2-1)\right]-\frac{1}{2}N\qquad(5)
\end{aligned}
$$

The difference between the number of dominance comparisons in case 3 and case 1 is calculated as follows.

$$
\begin{aligned}
\mathtt{dC}_{\mathtt{ss}-3}-\mathtt{dC}_{\mathtt{ss}-1}&=\frac{1}{3}X^3+\frac{1}{4}(4z+1)X^2-\frac{1}{12}\left[12zX-(6z^2-1)\right]X+\frac{1}{12}\left[(\delta^2-\delta)(2\delta+5)-6\delta(z^2-1)\right]-\frac{1}{2}N-\\
&\quad\left[\frac{1}{3}X^3+\frac{1}{4}X^2-\frac{1}{12}X-\frac{1}{2}N\right]\\
&=\frac{1}{3}X^3+\frac{1}{4}(4z+1)X^2-\frac{1}{12}\left[12zX-(6z^2-1)\right]X+\frac{1}{12}\left[(\delta^2-\delta)(2\delta+5)-6\delta(z^2-1)\right]-\frac{1}{2}N-\\
&\quad\frac{1}{3}X^3-\frac{1}{4}X^2+\frac{1}{12}X+\frac{1}{2}N
\end{aligned}
$$

$$
\begin{aligned}
&= zX^2 - \frac{1}{12}\left[12zX - (6z^2-1)\right]X + \frac{1}{12}\left[(\delta^2-\delta)(2\delta+5) - 6\delta(z^2-1)\right] + \frac{1}{12}X \\
&= zX^2 - zX^2 + \frac{1}{12}6z^2X - \frac{1}{12}X + \frac{1}{12}\left[(\delta^2-\delta)(2\delta+5) - 6\delta(z^2-1)\right] + \frac{1}{12}X \\
&= 6z^2X + \left[2\delta^3 - 2\delta^2 + 5\delta^2 - 5\delta - 6\delta z^2 + 6\delta\right] \\
&= 6z^2X + 2\delta^3 + 3\delta^2 + \delta - 6\delta z^2 \\
&= 6z^2(X-\delta) + 2\delta^3 + 3\delta^2 + \delta
\end{aligned} \tag{6}
$$

2 Best Case Analysis of ENS-BS

The number of dominance comparisons in case 1 is calculated using Eq. (7).

$$
\mathtt{dC}_{\mathtt{BS}-1} = \underbrace{\frac{1}{2}\sum_{k=1}^{K} n_k^2}_{A_{\mathtt{BS-1}}} + \underbrace{\sum_{k=1}^{K} \lceil \log k \rceil n_k}_{B_{\mathtt{BS-1}}} - \frac{1}{2}N \tag{7}
$$

Now we first compute $A_{\mathtt{BS-1}}$ and $B_{\mathtt{BS-1}}$.

$$
\begin{aligned}
A_{\mathtt{BS-1}} &= \frac{1}{2}\sum_{k=1}^{K} n_k^2 \\
&= \frac{1}{2}\sum_{k=1}^{2^y} (\lambda - \lceil \log k \rceil)^2 \\
&= \frac{1}{2}\left[\lambda^2 + (\lambda-1)^2 + 2(\lambda-2)^2 + 2^2(\lambda-3)^2 + 2^3(\lambda-4)^2 + \cdots + 2^{y-1}(\lambda-y)^2\right] \\
&= \frac{1}{2}\lambda^2 + \frac{1}{2}\sum_{l=1}^{y} 2^{l-1}(\lambda-l)^2 \\
&= \frac{1}{2}x^2 + \frac{1}{2}\sum_{l=1}^{x-1} 2^{l-1}(x-l)^2
\end{aligned} \tag{8}
$$

$$
\begin{aligned}
B_{\mathtt{BS-1}} &= \sum_{k=1}^{K} \lceil \log k \rceil n_k \\
&= \sum_{k=1}^{2^y} \lceil \log k \rceil \left(\lambda - \lceil \log k \rceil\right) \\
&= 0\cdot\lambda + 1\cdot(\lambda-1) + 2\cdot 2\cdot(\lambda-2) + 3\cdot 2^2\cdot(\lambda-3) + 4\cdot 2^3\cdot(\lambda-4) + \cdots + y\cdot 2^{y-1}\cdot(\lambda-y) \\
&= \sum_{l=1}^{y} l2^{l-1}(\lambda-l) \\
&= \sum_{l=1}^{x-1} l2^{l-1}(x-l)
\end{aligned} \tag{9}
$$

Now put the values of $A_{\mathtt{BS-1}}$ and $B_{\mathtt{BS-1}}$ in Eq. (7) to obtain the number of dominance comparisons in case 1 using Eq. (10).

$$
\begin{aligned}
\mathtt{dC}_{\mathtt{BS}-1} &= A_{\mathtt{BS-1}} + B_{\mathtt{BS-1}} - \frac{1}{2}N \\
&= \frac{1}{2}x^2 + \frac{1}{2}\sum_{l=1}^{x-1} 2^{l-1}(x-l)^2 + \sum_{l=1}^{x-1} l2^{l-1}(x-l) - \frac{1}{2}N
\end{aligned}
$$

$$= \frac{1}{2}x^2 + \frac{1}{2}\sum_{l=1}^{x-1} 2^{l-1}(x^2 + l^2 - 2xl) + \sum_{l=1}^{x-1} 2^{l-1}(xl - l^2) - \frac{1}{2}N$$
$$= \frac{1}{2}x^2 + \frac{1}{2}x^2\sum_{l=1}^{x-1} 2^{l-1} + \frac{1}{2}\sum_{l=1}^{x-1} l^2 2^{l-1} - x\sum_{l=1}^{x-1} l2^{l-1} + x\sum_{l=1}^{x-1} 2^{l-1}l - \sum_{l=1}^{x-1} l^2 2^{l-1} - \frac{1}{2}N$$
$$= \frac{1}{2}x^2 + \frac{1}{2}x^2(2^{x-1} - 1) - \frac{1}{2}\sum_{l=1}^{x-1} l^2 2^{l-1} - \frac{1}{2}N$$
$$= \frac{1}{2}x^2 + \frac{1}{2}x^2 2^{x-1} - \frac{1}{2}x^2 - \frac{1}{2}\sum_{l=1}^{x-1} l^2 2^{l-1} - \frac{1}{2}N$$
$$= x^2 2^{x-2} - \frac{1}{2}\sum_{l=1}^{x-1} l^2 2^{l-1} - \frac{1}{2}N \tag{10}$$

The number of dominance comparisons in case 3 is calculated using Eq. (11).

$$\mathtt{dC}_{\mathtt{BS-3}} = \underbrace{\frac{1}{2}\sum_{k=1}^{K} n_k^2}_{A_{\mathtt{BS-3}}} + \underbrace{\sum_{k=1}^{K} \lceil \log k \rceil n_k}_{B_{\mathtt{BS-3}}} - \frac{1}{2}N \tag{11}$$

Now we first compute $A_{\mathtt{BS-3}}$ and $B_{\mathtt{BS-3}}$.

$$A_{\mathtt{BS-3}} = \frac{1}{2}\sum_{k=1}^{K} n_k^2$$
$$= \frac{1}{2}\sum_{k=1}^{2^y+z} (\lambda - \lceil \log k \rceil)^2$$
$$= \frac{1}{2}\left[\lambda^2 + (\lambda-1)^2 + 2(\lambda-2)^2 + 2^2(\lambda-3)^2 + 2^3(\lambda-4)^2 + \cdots + 2^{y-1}(\lambda-y)^2 + z(\lambda-y-1)^2\right]$$
$$= \frac{1}{2}\lambda^2 + \frac{1}{2}\sum_{l=1}^{y} 2^{l-1}(\lambda-l)^2 + \frac{1}{2}z(\lambda-y-1)^2 \tag{12}$$

$$B_{\mathtt{BS-3}} = \sum_{k=1}^{K} \lceil \log k \rceil n_k$$
$$= \sum_{k=1}^{2^y+z} \lceil \log k \rceil (\lambda - \lceil \log k \rceil)$$
$$= 0\cdot\lambda + 1\cdot(\lambda-1) + 2\cdot 2\cdot(\lambda-2) + 3\cdot 2^2\cdot(\lambda-3) + 4\cdot 2^3\cdot(\lambda-4) + \cdots +$$
$$y\cdot 2^{y-1}\cdot(\lambda-y) + (y+1)\cdot z\cdot(\lambda-y-1)$$
$$= \sum_{l=1}^{y} l2^{l-1}(\lambda-l) + (y+1)z(\lambda-y-1) \tag{13}$$

Now put the values of $A_{\mathtt{BS-3}}$ and $B_{\mathtt{BS-3}}$ in Eq. (11) to obtain the number of dominance comparisons in case 3 using Eq. (14).

$$\mathtt{dC}_{\mathtt{BS-3}} = A_{\mathtt{BS-3}} + B_{\mathtt{BS-3}} - \frac{1}{2}N$$
$$= \frac{1}{2}\lambda^2 + \frac{1}{2}\sum_{l=1}^{y} 2^{l-1}(\lambda-l)^2 + \frac{1}{2}z(\lambda-y-1)^2 + \sum_{l=1}^{y} l2^{l-1}(\lambda-l) + (y+1)z(\lambda-y-1) - \frac{1}{2}N$$

$$= \frac{1}{2}\lambda^2 + \frac{1}{2}\sum_{l=1}^{y} 2^{l-1}(\lambda - l)^2 + \sum_{l=1}^{y} l2^{l-1}(\lambda - l) + \frac{1}{2}z(\lambda - y - 1)^2 + (y+1)z(\lambda - y - 1) - \frac{1}{2}N$$

$$= \frac{1}{2}\lambda^2 + \frac{1}{2}\sum_{l=1}^{y} 2^{l-1}(\lambda - l)\left[(\lambda - l) + 2l\right] + \frac{1}{2}z(\lambda - y - 1)\left[(\lambda - y - 1) + 2(y+1)\right] - \frac{1}{2}N$$

$$= \frac{1}{2}\lambda^2 + \frac{1}{2}\sum_{l=1}^{y} 2^{l-1}(\lambda - l)(\lambda + l) + \frac{1}{2}z(\lambda - y - 1)(\lambda + y + 1) - \frac{1}{2}N$$

$$= \frac{1}{2}\lambda^2 + \frac{1}{2}\sum_{l=1}^{y} 2^{l-1}(\lambda^2 - l^2) + \frac{1}{2}z\left[\lambda^2 - (y+1)^2\right] - \frac{1}{2}N$$

$$= \frac{1}{2}\lambda^2 + \frac{1}{2}\lambda^2\sum_{l=1}^{y} 2^{l-1} - \frac{1}{2}\sum_{l=1}^{y} l^2 2^{l-1} + \frac{1}{2}z\left[\lambda^2 - (y+1)^2\right] - \frac{1}{2}N$$

$$= \frac{1}{2}\lambda^2 + \frac{1}{2}\lambda^2(2^y - 1) - \frac{1}{2}\sum_{l=1}^{y} l^2 2^{l-1} + \frac{1}{2}z\left[\lambda^2 - (y+1)^2\right] - \frac{1}{2}N$$

$$= \frac{1}{2}\lambda^2 2^y - \frac{1}{2}\sum_{l=1}^{y} l^2 2^{l-1} + \frac{1}{2}z\left[\lambda^2 - (y+1)^2\right] - \frac{1}{2}N \tag{14}$$

Now we obtain the difference between the number of dominance comparisons in case 3 and case 1. So we subtract Eq. (10) from Eq. (14).

$$\mathtt{dC}_{\mathtt{BS}-3} - \mathtt{dC}_{\mathtt{BS}-1} = \frac{1}{2}\lambda^2 2^y - \frac{1}{2}\sum_{l=1}^{y} l^2 2^{l-1} + \frac{1}{2}z\left[\lambda^2 - (y+1)^2\right] - \frac{1}{2}N - x^2 2^{x-2} + \frac{1}{2}\sum_{l=1}^{x-1} l^2 2^{l-1} + \frac{1}{2}N$$

$$= \frac{1}{2}\lambda^2 2^y - \frac{1}{2}\sum_{l=1}^{y} l^2 2^{l-1} + \frac{1}{2}z\left[\lambda^2 - (y+1)^2\right] - x^2 2^{x-2} + \frac{1}{2}\sum_{l=1}^{x-1} l^2 2^{l-1}$$

$$= \frac{1}{2}\lambda^2 2^y + \frac{1}{2}z\left[\lambda^2 - (y+1)^2\right] - x^2 2^{x-2} - \frac{1}{2}\sum_{l=1}^{x-\delta} l^2 2^{l-1} + \frac{1}{2}\sum_{l=1}^{x-\delta} l^2 2^{l-1} + \frac{1}{2}\sum_{l=x-\delta+1}^{x-1} l^2 2^{l-1}$$

$$= \frac{1}{2}\lambda^2 2^y + \frac{1}{2}z\left[\lambda^2 - (y+1)^2\right] - x^2 2^{x-2} + \underbrace{\frac{1}{2}\sum_{l=x-\delta+1}^{x-1} l^2 2^{l-1}}_{\gamma}$$

$$= \frac{1}{2}\lambda^2 2^y + \frac{1}{2}z\left[\lambda^2 - (y+1)^2\right] - x^2 2^{x-2} + \left[(x-2)^2 2^{x-2} + 2^{x-1}\right] - \left[\left\{x - (\delta+1)\right\}^2 2^{x-\delta-1} + 2^{x-\delta}\right]$$

$$= \frac{1}{2}\lambda^2 2^y + \frac{1}{2}z\left[\lambda^2 - (y+1)^2\right] - x^2 2^{x-2} + (x-2)^2 2^{x-2} + 2^{x-1} - \left\{x - (\delta+1)\right\}^2 2^{x-\delta-1} - 2^{x-\delta}$$

$$= \frac{1}{2}\lambda^2 2^y + \frac{1}{2}z\left[\lambda^2 - (y+1)^2\right] + 2^{x-2}\left[(x-2)^2 - x^2\right] + 2^{x-1} - \left\{x - (\delta+1)\right\}^2 2^{x-\delta-1} - 2^{x-\delta}$$

$$= \frac{1}{2}\lambda^2 2^y + \frac{1}{2}z\left[\lambda^2 - (y+1)^2\right] + 2^{x-2}\left[4 - 4x\right] + 2^{x-1} - \left\{x - (\delta+1)\right\}^2 2^{x-\delta-1} - 2^{x-\delta}$$

$$= \frac{1}{2}\lambda^2 2^y + \frac{1}{2}z\left[\lambda^2 - (y+1)^2\right] - 2^x(x-1) + 2^{x-1} - \left\{x - (\delta+1)\right\}^2 2^{x-\delta-1} - 2^{x-\delta}$$

$$= \lambda^2 2^{y-1} + \frac{1}{2}z(\lambda - y - 1)(\lambda - y + 1) - (2x-3)2^{x-1} - \left\{x - (\delta+1)\right\}^2 2^{x-\delta-1} - 2^{x-\delta}$$

$$= \lambda^2 2^{y-1} + \frac{1}{2}\left[2^x - 2^y - (\lambda - y)2^y\right](\lambda + y + 1) - (2x-3)2^{x-1} -$$
$$\left\{x - (\delta+1)\right\}^2 2^{x-\delta-1} - 2^{x-\delta} \quad \text{Replace the value of } z(\lambda - y - 1) \text{ from Eq. (40) in the paper}$$

$$= \lambda^2 2^{x-\delta-1} + \left[2^{x-1} - 2^{x-\delta-1} - (\lambda - y)2^{x-\delta-1}\right](\lambda + y + 1) - (2x-3)2^{x-1} -$$
$$\left\{x - (\delta+1)\right\}^2 2^{x-\delta-1} - 2^{x-\delta}$$

$$
\begin{aligned}
&= \lambda^2 2^{x-\delta-1} + \lambda 2^{x-1} - \lambda 2^{x-\delta-1} - \lambda(\lambda-y)2^{x-\delta-1} + y2^{x-1} - y2^{x-\delta-1} - y(\lambda-y)2^{y-1} + \\
&\quad 2^{x-1} - 2^{x-\delta-1} - (\lambda-y)2^{x-\delta-1} - (2x-3)2^{x-1} - \left\{x-(\delta+1)\right\}^2 2^{x-\delta-1} - 2^{x-\delta} \\
&= (\lambda+y+1-2x+3)2^{x-1} + \left[\lambda^2 - \lambda - \lambda(\lambda-y) - y - y(\lambda-y) - 1 - (\lambda-y) - \right. \\
&\quad \left. \left\{x-(\delta+1)^2\right\} - 2\right] 2^{x-\delta-1} \\
&= (\lambda+x-\delta+1-2x+3)2^{x-1} + \left[\lambda^2 - \lambda - \lambda^2 + \lambda y - y - \lambda y + y^2 - \lambda + y - \right. \\
&\quad \left. \left\{x^2+\delta^2+2\delta+1-2x\delta-2x\right\} - 3\right] 2^{x-\delta-1} \\
&= (\lambda-x-\delta+4)2^{x-1} + \left[-2\lambda + y^2 - x^2 - \delta^2 - 2\delta - 1 + 2x\delta + 2x - 3\right] 2^{x-\delta-1} \\
&= (\lambda-x-\delta+4)2^{x-1} + \left[-2\lambda + (x-\delta)^2 - x^2 - \delta^2 - 2\delta - 1 + 2x\delta + 2x - 3\right] 2^{x-\delta-1} \\
&= (\lambda-x-\delta+4)2^{x-1} + \left[-2\lambda + x^2 + \delta^2 - 2x\delta - x^2 - \delta^2 - 2\delta - 1 + 2x\delta + 2x - 3\right] 2^{x-\delta-1} \\
&= (\lambda-x-\delta+4)2^{x-1} + \left[-2\lambda - 2\delta + 2x - 4\right] 2^{x-\delta-1} \\
&= (\lambda-x-\delta+4)2^{x-1} + \left[-\lambda - \delta + x - 2\right] 2^{x-\delta} \\
&= (\lambda-x-\delta+4)2^{x-1} - (\lambda-x+\delta+2)2^{x-\delta} \\
&= 2^{x-\delta}\left[(\lambda-x-\delta+4)2^{\delta-1} - (\lambda-x+\delta+2)\right] \\
&= 2^{x-\delta}\left[\left\{(\lambda+4)-(x+\delta)\right\}2^{\delta-1} - (\lambda-x+\delta+2)\right] \qquad (15)
\end{aligned}
$$

$$
\begin{aligned}
\gamma &= \frac{1}{2}\sum_{l=x-\delta+1}^{x-1} l^2 2^{l-1} \\
&= \frac{1}{2}\sum_{l=1}^{x-1} l^2 2^{l-1} - \frac{1}{2}\sum_{l=1}^{x-\delta} l^2 2^{l-1} \\
&= \frac{1}{2}\left[(x-1)^2 2^{x-1} + 2^x - (2x-3)2^{x-1} - 3\right] - \frac{1}{2}\left[(x-\delta)^2 2^{x-\delta} + 2^{x-\delta+1} - (2(x-\delta)-1)2^{x-\delta} - 3\right] \\
&= \left[(x-1)^2 2^{x-2} + 2^{x-1} - (2x-3)2^{x-2}\right] - \left[(x-\delta)^2 2^{x-\delta-1} + 2^{x-\delta} - (2(x-\delta)-1)2^{x-\delta-1}\right] \\
&= \left[2^{x-2}\left\{(x-1)^2 - (2x-3)\right\} + 2^{x-1}\right] - \left[2^{x-\delta-1}\left\{(x-\delta)^2 - (2(x-\delta)-1)\right\} + 2^{x-\delta}\right] \\
&= \left[2^{x-2}\left\{x^2+1-2x-2x+3\right\} + 2^{x-1}\right] - \left[2^{x-\delta-1}\left\{x^2+\delta^2-2x\delta-2x+2\delta+1\right\} + 2^{x-\delta}\right] \\
&= \left[2^{x-2}\left\{x^2+1-2x-2x+3\right\} + 2^{x-1}\right] - \left[2^{x-\delta-1}\left\{x^2+\delta^2-2x\delta-2x+2\delta+1\right\} + 2^{x-\delta}\right] \\
&= \left[2^{x-2}\left\{x^2-4x+4\right\} + 2^{x-1}\right] - \left[2^{x-\delta-1}\left\{x^2-2x(\delta+1)+\delta^2+2\delta+1\right\} + 2^{x-\delta}\right] \\
&= \left[2^{x-2}(x-2)^2 + 2^{x-1}\right] - \left[2^{x-\delta-1}\left\{x-(\delta+1)\right\}^2 + 2^{x-\delta}\right] \\
&= \left[(x-2)^2 2^{x-2} + 2^{x-1}\right] - \left[\left\{x-(\delta+1)\right\}^2 2^{x-\delta-1} + 2^{x-\delta}\right] \qquad (16)
\end{aligned}
$$